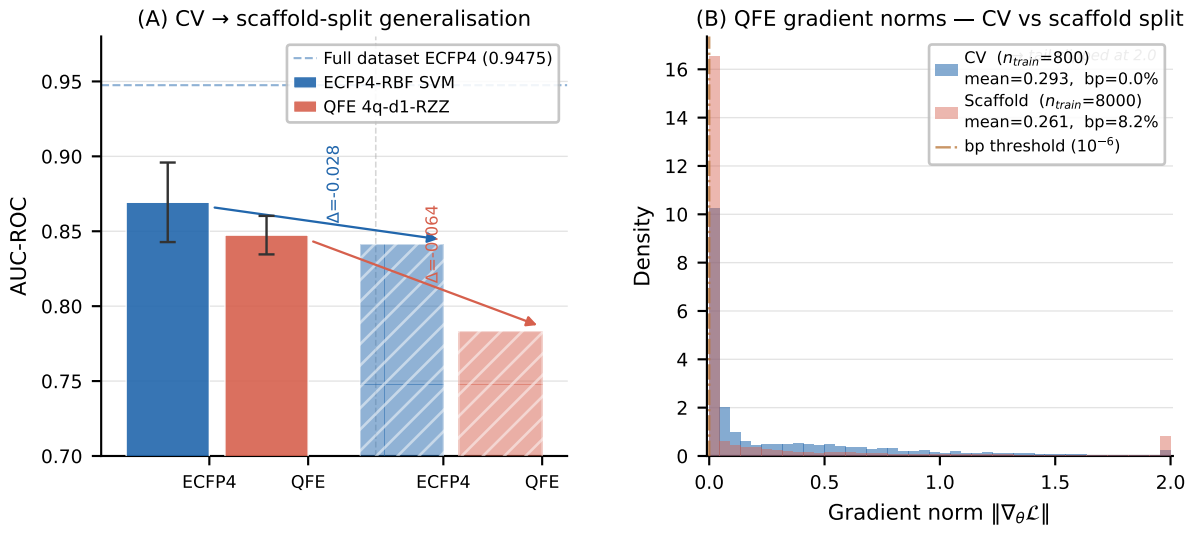
(A) CV → scaffold-split generalisation
Full dataset ECFP4 (0.9475)
ECFP4-RBF SVM
QFE 4q-d1-RZZ
Δ=-0.028
Δ=-0.064
AUC-ROC
0.95
0.90
0.85
0.80
0.75
0.70
ECFP4
QFE
ECFP4
QFE
(B) QFE gradient norms — CV vs scaffold split
CV (n_{train}=800)
mean=0.293, bp=0.0%
Scaffold (n_{train}=8000)
mean=0.261, bp=8.2%
bp threshold (10^{-6})
Density
16
14
12
10
8
6
4
2
0
0.0
0.5
1.0
1.5
2.0
Gradient norm $\|\nabla_\theta \mathcal{L}\|$